NHSAT–Algiers, Math. & CS
1st Cycle, 2nd Year
2025/2026

Geometrical and Wave Optics
Ms. CHEGGOU

Introduction To Optic-Fibers And Data Loss Related To The Bend Radius

1 Introduction to Fiber-Optics

Optical fibers work by confining and guiding the light wave within a long strand of glass by the principle of total internal reflection. Which are cylindrical dielectric waveguides made of a central cylinder of glass with one index of refraction, surrounded by an annulus with slightly different index of refraction . If the refractive indices of the core and cladding are n_1 and n_2 respectively, then for a ray entering the fiber, if the angle of incident (at the core cladding interface) θ_A, is greater than the critical angle, θ_C , then the ray will undergo total internal reflection at that interface.

2 Bending Loss

Bending loss in optical fibers includes micro and macro bending. Micro bend loss arises from microscopic deformations at the core-cladding interface, often due to poor design and installation, leading to non-uniform stresses. Macro bend loss occurs when the fiber experiences bends with a large radius of curvature, typically during installation, which affects the angle of incidence and disrupts total internal reflection, causing light to escape the core. Both types significantly contribute to overall light loss during fiber propagation.

3 Equations used

To calculate the bending loss we use the following formulas:

$$\theta_C = \sin^{-1}(\frac{n_2}{n_1}) \tag{1}$$

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \approx \frac{n_1 - n_2}{n_1} \tag{2}$$

Bend loss formula:

$$L(R) = \eta_{R1} e^{-\eta_{R1} R} \tag{3}$$

$L(R)$ = bend loss in **dB per turn**
R = bend radius in **mm**
$\eta_{R1} = 70$
$\eta_{R2} = 0.5\ \mathrm{mm}^{-1}$
dB to Linear Power Conversion: $P_{\text{transmitted}} = 10^{-L/10}$
Escape Probability: $P_{\text{escape}} = 1 - P_{\text{transmitted}}$
Data Transfer Percentage: $\text{Transfer}(R) = 100 \times 10^{-\frac{70e^{-0.5R}}{10}}$

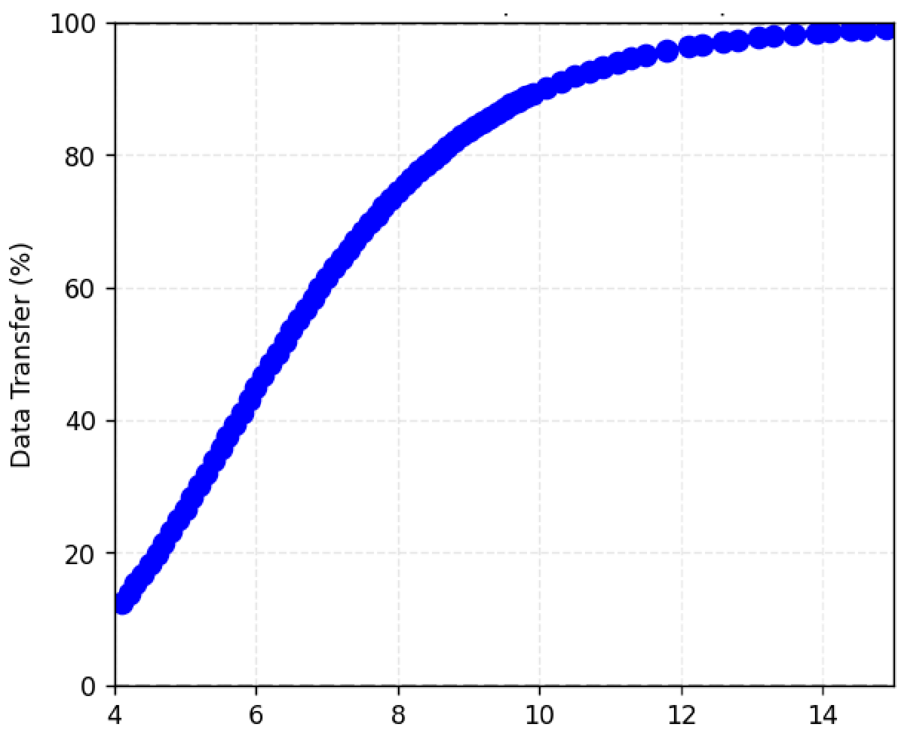

Figure 1: Bend Radius (mm)figure showing the python script output

4 Error of margin:

The bend loss is computed using an empirical exponential model with an expected uncertainty of approximately ±20% to ±40% in loss (dB) depending on fiber parameters, wavelength, and bend geometry.